## C3.9 Snoop channel dependencies

There are dependencies between the signals on different snoop channels.

In Figure C3-1:

- single-headed arrows point to signals that can be asserted before or after the signal at the start of the arrow
- double-headed arrows point to signals that must be asserted only after assertion of the signal at the start of the arrow.

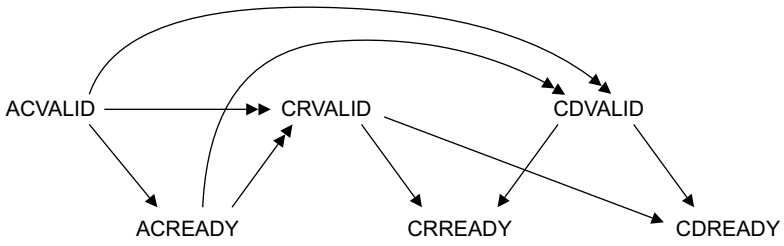



**Figure C3-1 Snoop channel dependencies**

Figure C3-1 shows the snoop address, snoop response, and snoop data channel dependencies and shows that:

- the interconnect must not wait for the master to assert **ACREADY** before asserting **ACVALID**
- the master can wait for **ACVALID** to be asserted before asserting **ACREADY**
- the master must wait for both **ACVALID** and **ACREADY** to be asserted before asserting **CRVALID**
- the master must wait for both **ACVALID** and **ACREADY** to be asserted before asserting **CDVALID**
- the master must not wait for the interconnect to assert **CRREADY** or **CDREADY** before asserting **CRVALID**
- if data transfer is required to complete the snoop operation, the master must not wait for the interconnect to assert **CRREADY** or **CDREADY** before asserting **CDVALID**
- the interconnect can wait for **CRVALID** or **CDVALID**, if data transfer is required, or both before asserting **CRREADY**
- the interconnect can wait for **CRVALID** or **CDVALID**, if data transfer is required, or both before asserting **CDREADY**.


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# Chapter C4
# Coherency Transactions on the Read Address and Write Address Channels

This chapter describes the transactions that can be issued by an initiating master on the read address and write address channels. The expected channel activity for each transaction group is described, and a brief overview is given for each transaction together with a description of the associated cache line state changes. It contains the following sections:


- *About an initiating master* on page C4-184
- *About snoop filtering* on page C4-187
- *State changes on different transactions* on page C4-188
- *State change descriptions* on page C4-190
- *Read transactions* on page C4-191
- *Clean transactions* on page C4-197
- *Make transactions* on page C4-200
- *Write transactions* on page C4-202
- *Evict transactions* on page C4-206
- *Handling overlapping write transactions* on page C4-207.



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